POLSCI 9592

Lecture 4: Model Fit and Evaluation

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Goals for This Session

1. Absolute Measures of Model Fit
2. Comparative Measures of Model Fit
3. Nested and Non-nested Model Tests
4. Model Specification Tests
5. Model Diagnostics
6. (Quasi-)Separation

Model Fit, Evaluation and Comparison

Now that we know how to interpret what is going on in these models, we need to develop a sense of how well they fit. We can do this through a number of different means.

- Likelihood-based fit measures
- Pseudo- R^2
- Information criteria
- Classification-based measures.
- Cross-validation.
- Specification tests.

Pseudo R-squared

Pseudo- R^2 measures rely on analogues to the linear model.

- Many different types, each of which can produce substantively different results from the others.
- If you use these measures, identify which one(s) you are using.
- None can be interpreted as the proportion of variation in the dependent variable explained by the independent variables.

McFadden's R-squared

McFadden's R^2 uses the following analogue to the linear model $R^2 = 1 - \frac{SS_{\text{residual}}}{SS_{\text{total}}}$. In this case,

$$\text{McFadden's } R^2 = 1 - \frac{\log \hat{L}(M_{\text{Full}})}{\log \hat{L}(M_{\text{Null}})}$$

There is also an adjustment to McFadden's R^2 to account for degrees of freedom:

$$\text{McFadden's } \bar{R}^2 = 1 - \frac{\log \hat{L}(M_{\text{Full}}) - K^*}{\log \hat{L}(M_{\text{Null}})}$$

where K^* is the number of parameters (not independent variables) in the model.

Cox & Snell and Cragg & Uhler R-squared

The Cox & Snell (or ML) R^2 uses the same analogy as McFadden's:

$$\text{Cox \& Snell } R^2 = 1 - \left\{ \frac{\hat{L}(M_{\text{Null}})}{\hat{L}(M_{\text{Full}})} \right\}^{\frac{2}{N}}$$

The Cox & Snell measure reaches a maximum of $1 - L(M_{\text{Null}})^{\frac{2}{N}}$, so Cragg and Uhler's R^2 norms it to reach a maximum at 1

$$\text{Cragg \& Uhler's } R^2 = \frac{1 - \left\{ \frac{\hat{L}(M_{\text{Null}})}{\hat{L}(M_{\text{Full}})} \right\}^{\frac{2}{N}}}{1 - L(M_{\text{Null}})^{\frac{2}{N}}}$$

Efron's R-squared

Efron's analogy to the linear model formula mentioned above is even more explicit.

$$\text{Efron's } R^2 = 1 - \frac{\sum_N (y_i - \hat{\pi}_i)^2}{\sum_N (y_i - \bar{y})^2}$$

McKelvey & Zavoina's R-squared

For models that can be defined in terms of a latent variable y^*: $y^* = \mathbf{x}\beta + \varepsilon$ where $\widehat{\mathrm{Var}}(\hat{y}^*) = \hat{\beta}' \widehat{\mathrm{Var}}(\mathbf{x}) \hat{\beta}$, The M&Z R^2 is:

$$\text{M\&Z } R^2 = \frac{\widehat{\mathrm{Var}}(\hat{y}^*)}{\widehat{\mathrm{Var}}(y^*)} = \frac{\widehat{\mathrm{Var}}(\hat{y}^*)}{\widehat{\mathrm{Var}}(\hat{y}^*) + \mathrm{Var}(\varepsilon)}$$

Here, $\mathrm{Var}(\varepsilon)$ identifies the model. In logit it is $\frac{\pi^2}{3}$ and in the probit model it is 1.

Measures and Models

	OLS	Binary	Ordered	Multinomial	Count
Log-Likelihood	✓	✓	✓	✓	✓
LR χ^2	✓	✓	✓	✓	✓
Information Criteria	✓	✓	✓	✓	✓
R^2 and $\bar{R}^2$	✓				
Efron's R^2 and Tjur's D		✓			
McFadden, ML, C&U R^2		✓	✓	✓	✓
Count and Adjusted Count R^2		✓	✓	✓	
$Var(e)$, $Var(y^*)$, M&Z R^2		✓	✓		
PRE and ePRE		✓	✓	✓	
AIC	✓	✓	✓	✓	✓
BIC	✓	✓	✓	✓	✓

Example

```
library(DAMisc)
load("data/anes_2008_binary.rda")
dat$race <- rio::factorize(dat$race)
mod <- glm(voted ~ age + educ + income + poly(leftright, 2) +
    female + race, data=dat, family=binomial(link="logit"))
binfit(mod)
```

```
##                     Names1    vals1                        Names2    vals2
## 1 Log-Lik Intercept Only: -314.550          Log-Lik Full Model: -249.803
## 2                 D(575):  499.606                       LR(8):  129.495
## 3                                                  Prob > LR:    0.000
## 4          McFadden's R2:    0.206           McFadden's Adk R2:    0.177
## 5      ML (Cox-Snell) R2:    0.199 Cragg-Uhler (Nagelkerke) R2:    0.302
## 6  McKelvey & Zavoina R2:    0.357                  Efron's R2:    0.214
## 7               Count R2:    0.795              Adj Count R2:    0.104
## 8                    BIC:  556.935                       AIC:  517.606
```

Classification-based Measures

Proportional Reduction in Error. (PRE) tells us how much better we do at predicting y_i using a model versus guessing.

$$PRE = \frac{PCP - PMC}{1 - PMC}$$

Here,

$$\hat{y}_i = \begin{cases} 1 & \text{if } \hat{p}_i > 0.5 \\ 0 & \text{if } \hat{p}_i \leq .5 \end{cases}$$

$$PMC = max\left(\frac{\sum_i y_i}{N}, \frac{\sum_i (1 - y_i)}{N}\right)$$

$$PCP = \frac{\#(y_i = \hat{y}_i)}{N}$$

Expected PRE

Herron (1999) shows that there are lots of different arrangements of predicted probabilities that would generate the same PRE: For example, consider the two different sets of predictions:

y_i	$\hat{p}_i$	$\hat{y}_i$
1	0.60	1
0	0.51	1
1	0.55	1
1	0.75	1
0	0.25	0
0	0.45	0
0	0.40	0

$$PMC = 4/7 = 0.57$$
$$PCP = 6/7 = 0.86$$

y_i	$\hat{p}_i$	$\hat{y}_i$
1	0.80	1
0	0.51	1
1	0.65	1
1	0.95	1
0	0.15	0
0	0.25	0
0	0.10	0

$$PMC = 4/7 = 0.57$$
$$PCP = 6/7 = 0.86$$

ePRE

The ePRE is defined as:

$$ePRE = \frac{ePCP - ePMC}{1 - ePMC}$$

where:

$$ePMC = \bar{y}$$

$$ePCP = \frac{1}{N} \sum_i (y_i \hat{p}_i + (1 - y_i)(1 - \hat{p}_i))$$

Expected PRE Example

y_i	$\hat{p}_i$	$\hat{y}_i$
1	0.60	1
0	0.51	1
1	0.55	1
1	0.75	1
0	0.25	0
0	0.45	0
0	0.40	0

$$ePMC = 0.429$$

$$ePCP = 0.613$$

$$ePRE = 0.323$$

y_i	$\hat{p}_i$	$\hat{y}_i$
1	0.80	1
0	0.51	1
1	0.65	1
1	0.95	1
0	0.15	0
0	0.25	0
0	0.10	0

$$ePMC = 0.429$$

$$ePCP = 0.77$$

$$ePRE = 0.598$$

PRE in R

```
pre(mod, sim=T)
```

```
## mod1:  voted ~ age + educ + income + poly(leftright, 2) + female + race
## mod2:  voted ~ 1
##
## Analytical Results
##  PMC =  0.771
##  PCP =  0.795
##  PRE =  0.104
## ePMC =  0.646
## ePCP =  0.724
## ePRE =  0.220
##
## Simulated Results
##      median lower upper
##  PRE 0.104  0.052 0.157
## ePRE 0.216  0.152 0.272
```

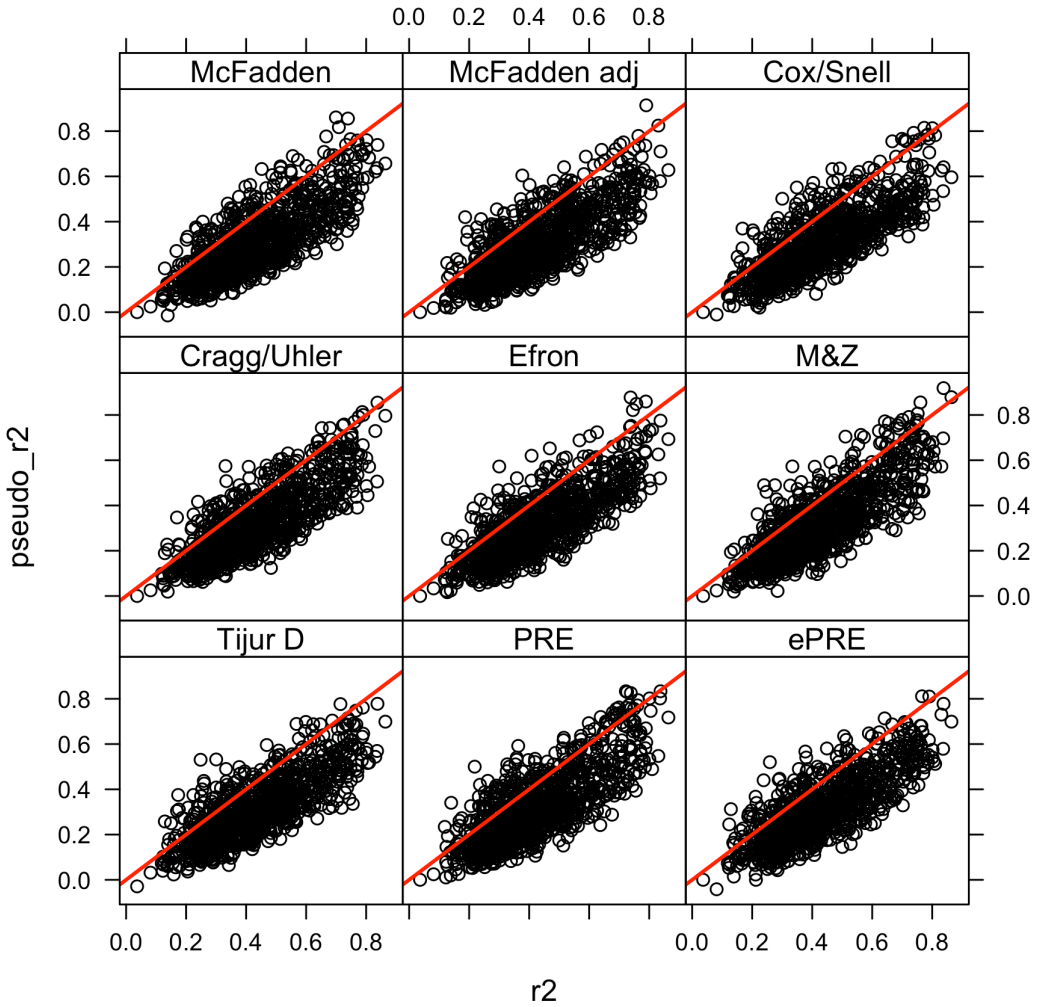
Others...

There are two other methods that consider the $Pr(Y = 1|Y = 1, X)$ and $1 - Pr(Y = 1|Y = 0, X)$.

- Separation Plot (Greenhill, Ward and Sacks, 2011) is a visual method that plots a single tick for each observation (where ones and zeros are different colors) sorted by $Pr(Y = 1|X)$.
- Tjur's D is a statistic that is defined as

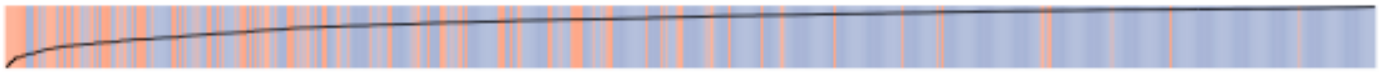
$$\frac{1}{n_1} \sum_{y=1} Pr(Y = 1|X) - \frac{1}{n_0} \sum_{y=0} Pr(Y = 1|X)$$

Comparison of Pseudo R-squared Measures



Separation Plot

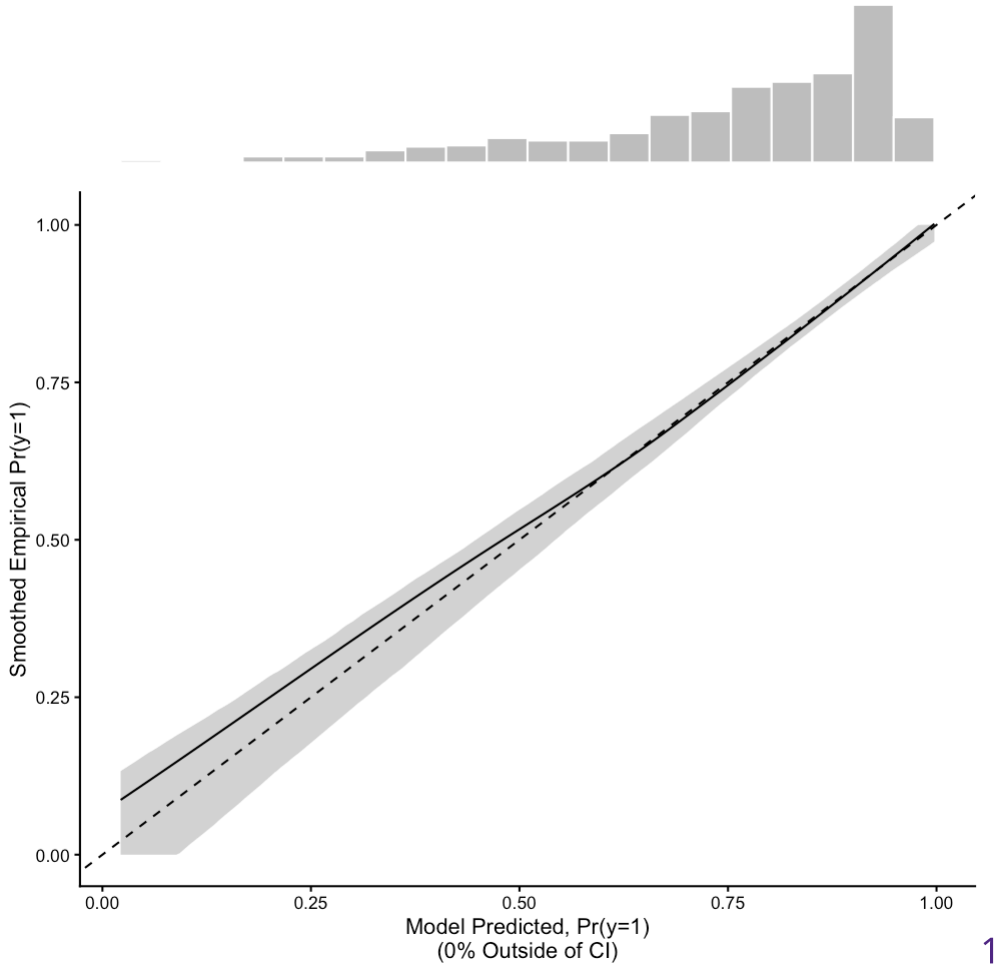
```
cols <- brewer.pal(5, "Set2")[c(2,3)]
y <- model.response(model.frame(mod))
separationplot(fitted(mod), c(y),
    col0=cols[1], col1=cols[2], file=NULL)
```



Empirical vs Predicted Probabilities

```
# remotes::install_github("davidaarmstrong/psre")
library(psre)
gh1 <- gg_hmf(
  model.response(model.frame(mod)),
  fitted(mod),
  method="loess")
```

```
library(patchwork)
gh1[[1]] + gh1[[2]] +
  plot_layout(heights=c(2,8), ncol=1)
```



Comparing Two Models: Nested

- Nested: use likelihood ratio test.

```
library(lmtest)
m1 <-  glm(voted ~ age + educ + income + poly(leftright, 2) +
    female + race, data=dat, family=binomial(link="logit"))
m2 <-  glm(voted ~ educ + income + poly(leftright, 2),
           data=dat, family=binomial(link="logit"))
lrtest(m1, m2)
```

```
## Likelihood ratio test
##
## Model 1: voted ~ age + educ + income + poly(leftright, 2) + female + race
## Model 2: voted ~ educ + income + poly(leftright, 2)
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   9 -249.80
## 2   5 -277.46 -4 55.311  2.797e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comparing Non-nested Models: AIC

Let's imagine that all models we are comparing are trying to explain the same objective reality (call it f).

- f is a function of ξ (the infinitely large set of all possible explanatory variables) and is non-parametric.

Further, imagine that we are estimating some model $g(\mathbf{X}|\theta)$ that predicts reality using $\mathbf{X}$, a subset of the variables in ξ and θ a set of parameters relating $\mathbf{X}$ to y. Then ...

$$\begin{aligned} I(f,g) &= E_f[\log(f(x))] - E_f[\log(g(x|\theta))] \\ &= C - E_f[\log(g(x|\theta))] \end{aligned}$$

Is the amount of information we lose when trying to approximate f with $g(\mathbf{X},\theta)$.

AIC

Akaike found that the $LL(g)$, the log-likelihood of the model was a biased estimator of $I(f, g)$, but the bias was on the order of K (the number of parameters in the model). Therefore:

$$\widehat{I(f, g)} = LL(\theta|\mathbf{X}, y) - K$$

To increase similarity to the already well-established likelihood ratio statistic, Akaike multiplied his measure by -2:

$$AIC = -2LL(\theta|\mathbf{X}, y) + 2K$$

Small Sample Correction

There is a small sample correction that should be used when n is small in the absolute terms or n is small relative to K (e.g., $\frac{n}{K} \leq 40$)

$$AIC_c = -2LL(\theta|\text{data}) + 2K + \frac{2K(K+1)}{n-K-1}$$

Comparing Non-nested Models: BIC

BIC is meant to approximate the Bayes Factor:

$$\frac{\Pr(D|M_1)}{\Pr(D|M_2)} = \frac{\int \Pr(\theta_1|M_1)\Pr(D|\theta_1, M_1)d\theta_1}{\int \Pr(\theta_2|M_2)\Pr(D|\theta_2, M_2)d\theta_2}$$

The approximation is defined as:

$$BIC = -2LL(\theta|\mathbf{X}, y) + K\log(n)$$

Delta values

Often, for AIC_c, AIC or BIC to be interpretable, Δ_i should be calculated such that for each model i in the model set,

$$\Delta_i = \star IC_i - \star IC_{\min}$$

This gives the "best" model $\Delta_i = 0$

- For AIC, this captures the additional expected information loss incurred by using model g_i rather than the best-supported approximating model in the candidate set, g_{min}.
  - The large Δ_i, the less support model i receives as the best approximating model in the candidate set.
- For BIC, Δ_i measures the decrease in approximate posterior support for model i relative to the model with the smallest BIC.
  - Larger values indicate that model i is increasingly less supported by the data relative to the model with the smallest BIC.

Interpreting Δ values

Conventional cut-off values for AIC Δ_i are:

- $\Delta_i \leq 2$ indicates substantial support,
- $4 \leq \Delta_i \leq 7$ indicates less support,
- $\Delta_i \geq 10$ indicates essentially no support.

For BIC, the Δ values translate directly into odds ratios of support for the minimum BIC model relative to the candidate model:

$$OR = e^{\left(\frac{\Delta}{2}\right)}$$

- $\Delta = 1.39$ has $OR \approx 2$
- $\Delta = 4.6$ has $OR \approx 10$
- $\Delta = 6$ has $OR \approx 20$
- $\Delta = 9.2$ has $OR \approx 100$

IC Weights

We can construct weights from the Δ_i values in the following way:

$$w_i = \frac{exp\left(\frac{-\Delta_i}{2}\right)}{\sum_i exp\left(\frac{-\Delta_i}{2}\right)}$$

- AIC w_i gives (essentially) the relative probability (or evidence) that model i is the KL-best approximating model among the candidate models.
- BIC w_i approximates the posterior probability among the candidate models, conditional on: the candidate model set, equal prior model probabilities, the Laplace approximation or the marginal likelihood.

AIC or BIC?

The question of whether to use AIC or BIC is often left to how much you want to penalize additional model parameters. In actuality, the question is one of performance in picking the K-L best model.

- When there are "tapering effects" that should remain in selected models, AIC is better
- When the candidate set contains a relatively simple model with most of the explanatory power, BIC often performs better.

It's worth noting, as a selection tool BIC's consistency in picking the best model depends on the "true" model being in the candidate set.

Comparing Non-nested Models: Clarke's Test

Clarke (2003) puts forth a distribution-free test that is really a "paired sign test". The statistic is calculated as:

$$d_i = \log(\mathcal{L}_{\beta,x_i}) - \log(\mathcal{L}_{\gamma,z_i}) - (p-q)\left(\frac{log(n)}{2n}\right)$$

$$B = \sum_{i=1}^{n} I_{0,+\infty}(d_i)$$

- The d_i are the difference in individual log-likelihoods for the two models
- The second equation above counts up the number of positive d_i values.
- We are testing to see whether B is significantly bigger than a random binomial variable that has a $p = .5$ and n the same as the number of rows in $\mathbf{X}$ and $\mathbf{Z}$.

Clarke Test in R

```
library(clarkeTest)
m1 <-  glm(voted ~ age + female + race, data=dat,
           family=binomial(link="logit"))
m2 <-  glm(voted ~ educ + income + poly(leftright, 2),
           data=dat, family=binomial(link="logit"))
IC_delta(m1, m2)
```

```
##     df D_AIC D_AICc D_BIC
## ..1  5 31.31  31.31 31.31
## ..2  5  0.00   0.00  0.00
```

```
clarke_test(m1, m2)
```

```
##
## Clarke test for non-nested models
##
## Model 1 log-likelihood: -293
## Model 2 log-likelihood: -277
## Observations: 584
## Test statistic: 246 (42%)
##
## Model 2 is preferred (p = 0.00016)
```

Comparing Non-nested Models: Cross-validation

```
library(rsample)
library(tidyr)
library(purrr)
dat$lrstren <- abs(dat$leftright-5)
m1 <-  glm(voted ~ age + female + race +
             lrstren, data=dat,
           family=binomial(link="logit"))
m2 <-  glm(voted ~ educ + income +
             poly(leftright, 2), data=dat,
           family=binomial(link="logit"))

cv_logit <- function(split, m1, m2, ...){
  m1_up <- update(m1, data=analysis(split))
  m2_up <- update(m2, data=analysis(split))
  y_out <- model.response(
            model.frame(formula(m1),
                        data=assessment(split)))
  p1 <- predict(m1_up,
                newdata=assessment(split),
                type="response")
  p2 <- predict(m2_up,
                newdata=assessment(split),
                type="response")
  ll1 <- sum(-y_out*log(p1) - (1-y_out)*log(1-p1))
  ll2 <- sum(-y_out*log(p2) - (1-y_out)*log(1-p2))
  tibble(model = factor(1:2,
                        labels=c("Model1", "Model2")),
         ll = c(ll1, ll2))
}
```

```
cv_out <- vfold_cv(dat,
                   v=10,
                   repeats=10) %>%
    mutate(ll = map(splits, cv_logit, m1, m2)) %>%
    unnest(ll) %>%
    group_by(id, model) %>%
    summarise(ll = sum(ll)) %>%
    pivot_wider(names_from="model", values_from="ll") %>%
  mutate(diff = Model2-Model1)
cv_out
```

```
## # A tibble: 10 × 4
## # Groups:   id [10]
##    id       Model1 Model2   diff
##    <chr>     <dbl>  <dbl>  <dbl>
##  1 Repeat01   296.   284. -11.6
##  2 Repeat02   298.   285. -12.9
##  3 Repeat03   294.   282. -12.2
##  4 Repeat04   296.   286.  -9.99
##  5 Repeat05   294.   283. -11.7
##  6 Repeat06   295.   284. -11.4
##  7 Repeat07   295.   285. -10.6
##  8 Repeat08   293.   282. -10.7
##  9 Repeat09   295.   283. -11.7
## 10 Repeat10   293.   283.  -9.31
```

Residuals, Outliers and Influential Observations

While looking at the residuals is marginally less interesting in these models, they can still tell us something about model fit.

- Many different kinds of residuals, all tell us something different about the model.

Response and Pearson Residuals

- Response residuals are: $y_i - \hat{p}_i$, but have little diagnostic value because they do not account for the inherent heteroskedasticity of the non-linear models.
- Pearson Residuals: $e_{Pi} = \frac{y_i - \hat{p}_i}{\sqrt{\text{var}(y_i|\mathbf{x})}} = \frac{y_i - \hat{p}_i}{\sqrt{\hat{p}_i(1-\hat{p}_i)}}$
- Standardized Pearson Residuals: $e_{PSi} = \frac{e_{Pi}}{\sqrt{1-h_i}}$ where $h_i = \hat{p}_i(1-\hat{p}_i)x_i\widehat{\text{Var}}\left(\hat{\beta}\right)x_i'$

Deviance Residuals

The deviance residuals are the observations contribution to the overall deviance:

$$e_{Di} = \text{sign}(y - \hat{p}_i)\sqrt{2}\sqrt{y_i \log\left(\frac{y_i}{\hat{p}_i}\right) + (1 - y_i)\log\left(\frac{1 - y_i}{1 - \hat{p}_i}\right)}$$

A standardized version of the deviance residual is also available:

$$e_{DSi} = \frac{e_{Di}}{\sqrt{1 - h_i}}$$

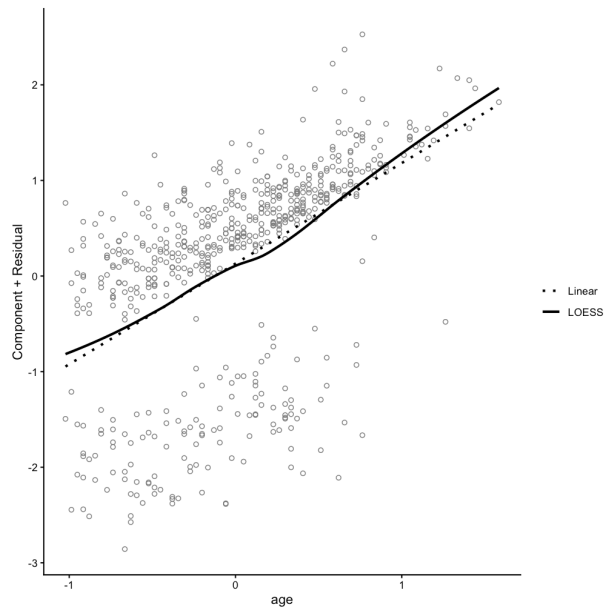
The standardized Pearson and deviance residuals can be obtained with `rstandard()` in R.

Component + Residual Plots

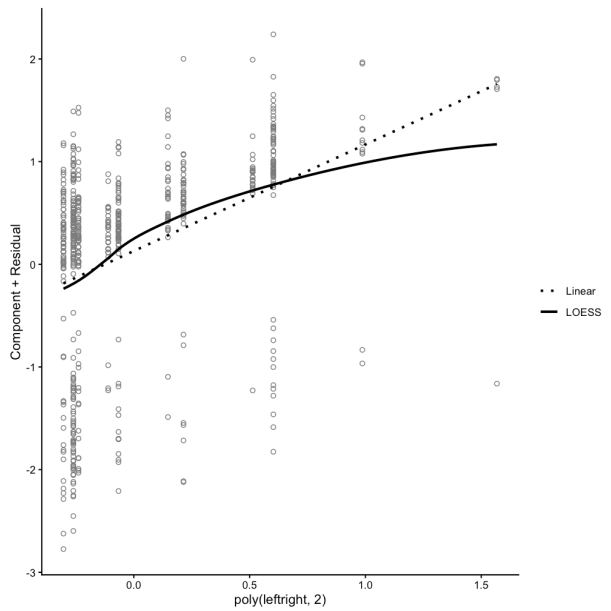
The `gg_crplot()` function is defined in the code for this class.

```
m1 <-  glm(voted ~ age + educ + income + poly(leftright, 2) +
    female + race, data=dat, family=binomial(link="logit"))
```

```
gg_crplot(m1, "age")
```




```
gg_crplot(m1, "poly(leftright, 2)")
```



New Model

```
dat$lrfac <- as.factor(dat$leftright)
mnew <- glm(voted ~ age + educ + income + as.factor(leftright) +
    female + race, data=dat, family=binomial(link="logit"))
anova(m1, mnew, test='Chisq')
```

```
## Analysis of Deviance Table
##
## Model 1: voted ~ age + educ + income + poly(leftright, 2) + female + race
## Model 2: voted ~ age + educ + income + as.factor(leftright) + female +
##     race
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       575     499.61
## 2       567     483.31  8     16.3  0.03828 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Outlier Diagnostics

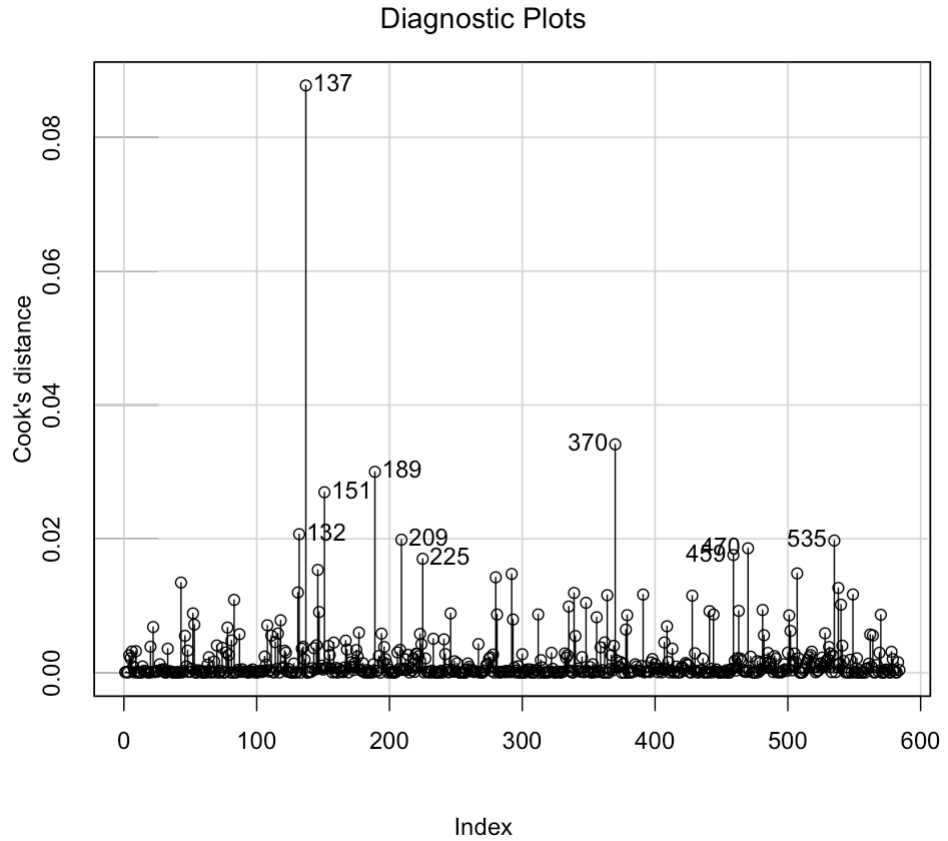
Cook's D can also be calculated for these models:

$$D_i = \frac{e^2_{PSi}}{K+1} \frac{h_i}{1-h_i}$$

The function `influenceIndexPlot` in R will produce an index plot of Cook's distances.

Residual Plots

```
library(car)
influenceIndexPlot(mod, vars="Cook", id=list(n=10))
```



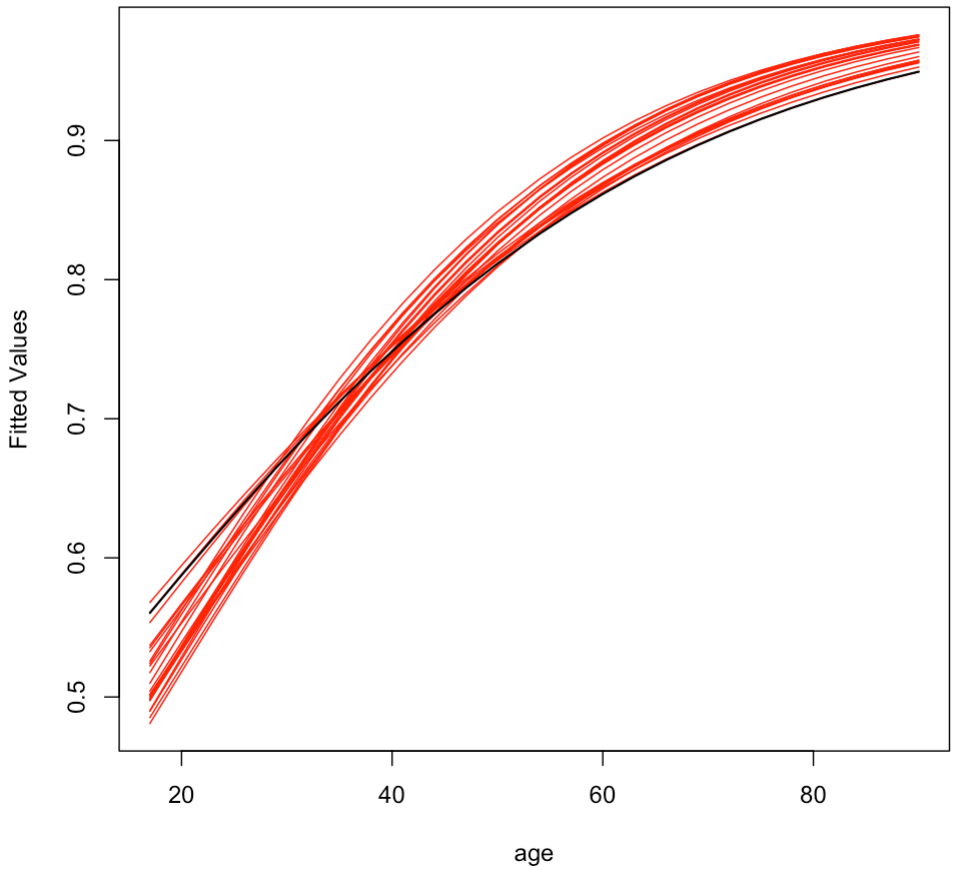
Looking at Deleted Obs Effects

I wrote a function called `outEff` that plots the effect of variables on predicted probabilities after removing observations that are thought to be most outlying.

- It can do this either one at a time or cumulatively.

outEff

```
outEff(mod,
       var='age',
       data=dat,
       nOut=25,
       cumulative=TRUE)
```



Separation/Small Sample Problems

MLE has a difficult time providing accurate (i.e., unbiased) estimates in the presence of small samples.

- Small samples in the traditional sense (overall small n).
- Small number of observations in the least populous category

Similar problems occur in a condition of separation (or quasi-separation). Zorn (2005) is a nice piece on this.

- No variability on the DV for some category of an independent variable.

Separation

The problem with separation is

- Parameter values try to take on values arbitrarily far away from zero (ultimately either $-\infty$ or ∞)

You know you have this problem if:

- You're using the `logit` command in Stata and it tells you that some observations were perfectly predicted and thus dropped.
- You're estimating a GLM in any software and you get huge coefficients with really huge standard errors.

Firth's Solution

Firth's solution is a penalized likelihood where:

- The penalty on the likelihood function drives parameters toward zero
- It does so most strongly for the parameters affected by separation, keeping their estimates finite.

The math is a bit beyond the scope of our discussion, but this is the accepted solution for separation problems across a wide range of disciplines.

Example: Voting for National Front

```
sepdat <- rio::import("data/france_binary.dta")
sepdat <- sepdat %>%
  mutate(across(c("demsat", "retnat", "union"),
                rio::factorize))
sepdat <- sepdat |>
  select(votefn, demsat, age, lrself,
         hhincome, retnat, union) |>
  na.omit()

mod3 <- glm(votefn ~ demsat + age + lrself + hhincome + retnat
    + union, data=sepdat, family=binomial)
printCoefmat(summary(mod3)$coefficients)
```

```
##                               Estimate  Std. Error z value Pr(>|z|)
## (Intercept)                -19.7125946 689.5448696 -0.0286   0.9772
## demsatSomewhat satisfied    15.2255675 689.5443893  0.0221   0.9824
## demsatA little satisfied    16.0828401 689.5444023  0.0233   0.9814
## demsatNot at all satisfied  16.8229674 689.5444325  0.0244   0.9805
## age                         -0.0039201   0.0084709 -0.4628   0.6435
## lrself                       0.2537232   0.0515159  4.9251 8.43e-07 ***
## hhincome                    -0.1305836   0.0860719 -1.5171   0.1292
## retnatsame                   0.5459725   0.6085426  0.8972   0.3696
## retnatworse                  0.8501427   0.5649549  1.5048   0.1324
## unionYes                    -0.3255606   0.5036256 -0.6464   0.5180
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Firth Logit

```
library(logistf)
mod3a <- logistf(votefn ~ demsat + age + lrself + hhincome + retnat
    + union, data=sepdat)
pfl(mod3a)
```

```
##                             coef    se(coef) p      lower 0.95 upper 0.95
## (Intercept)                -6.1860 1.5819   0.0000 -11.1977   -3.5833
## demsatSomewhat satisfied    1.8711 1.3854   0.0762  -0.1462    6.7233
## demsatA little satisfied    2.7207 1.3916   0.0036   0.6813    7.5774
## demsatNot at all satisfied  3.4529 1.4053   0.0001   1.3640    8.3195
## age                        -0.0037 0.0081   0.6573  -0.0203    0.0126
## lrself                      0.2483 0.0494   0.0000   0.1504    0.3499
## hhincome                   -0.1280 0.0820   0.1300  -0.2964    0.0375
## retnatsame                  0.4700 0.5603   0.4047  -0.6084    1.7154
## retnatworse                 0.7421 0.5196   0.1384  -0.2207    1.9295
## unionYes                   -0.2480 0.4669   0.5978  -1.3016    0.6217
##                            Chisq
## (Intercept)                32.4146
## demsatSomewhat satisfied    3.1445
## demsatA little satisfied    8.4590
## demsatNot at all satisfied 14.9172
## age                         0.1968
## lrself                     25.2481
## hhincome                    2.2929
## retnatsame                  0.6944
## retnatworse                 2.1958
## unionYes                    0.2783
```

Firth's Problem: Predicted Probabilities

Firth's penalty is equivalent to adding pseudo-observations to the data:

- Each observation is copied twice, once with $y = 1$ and once with $y = 0$, each with weight $h_i/2$ (where h_i is the observation's leverage).
- The pseudo-data are perfectly balanced, so they pull everything toward $Pr(y = 1) = 0.5$, including the intercept.

The result:

- The slope coefficients are good (less biased than ML).
- The predicted probabilities are biased toward 0.5. In ML logit, the average predicted probability equals the observed proportion of 1s; with Firth, it doesn't.
- The problem is worst when the outcome is rare (or very common), which is exactly when separation tends to happen.

Puhr et al. (2017) propose two fixes: FLIC and FLAC.

FLIC: Intercept Correction

FLIC (Firth Logistic regression with Intercept Correction)

- Keep the Firth slopes.
- Re-estimate only the intercept by ML, holding the Firth linear predictor (without the intercept) fixed as an offset.
- The average predicted probability now equals the observed proportion of 1s.
- Fit with `logistf(..., flic=TRUE)` or `flic(mod)`.

Note: the CI that `logistf` reports for the FLIC intercept is a Wald interval and ignores uncertainty in the slopes.

FLAC: Added Covariate

FLAC (Firth Logistic regression with Added Covariate)

- Fit ML to the augmented data (original + pseudo-observations), adding a dummy variable that identifies the pseudo-observations.
- The dummy soaks up the pseudo-data's pull toward 0.5; predictions set it to 0.
- The average predicted probability again equals the observed proportion of 1s.
- Fit with `flac(mod, data)`. Profile likelihood CIs are available for all coefficients.

FLIC keeps the Firth slopes exactly; FLAC's slopes can differ somewhat. Both matter most for predicted probabilities, much less for first differences, where an intercept shift mostly cancels out.

Profile Likelihood Confidence Intervals

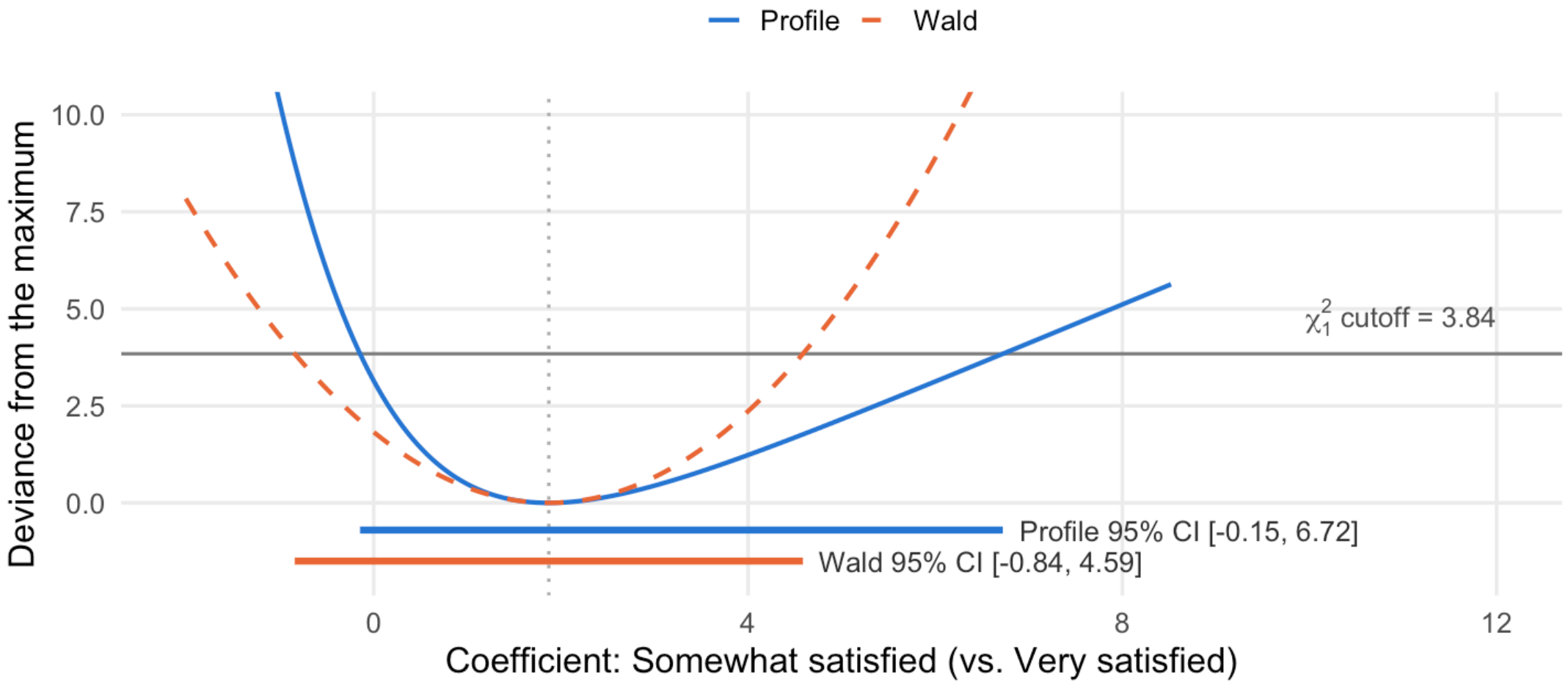
Wald intervals, $\hat{\beta} \pm 1.96 \times SE$, assume the log-likelihood is a parabola around $\hat{\beta}$, so they are always symmetric.

Profile intervals use the likelihood itself:

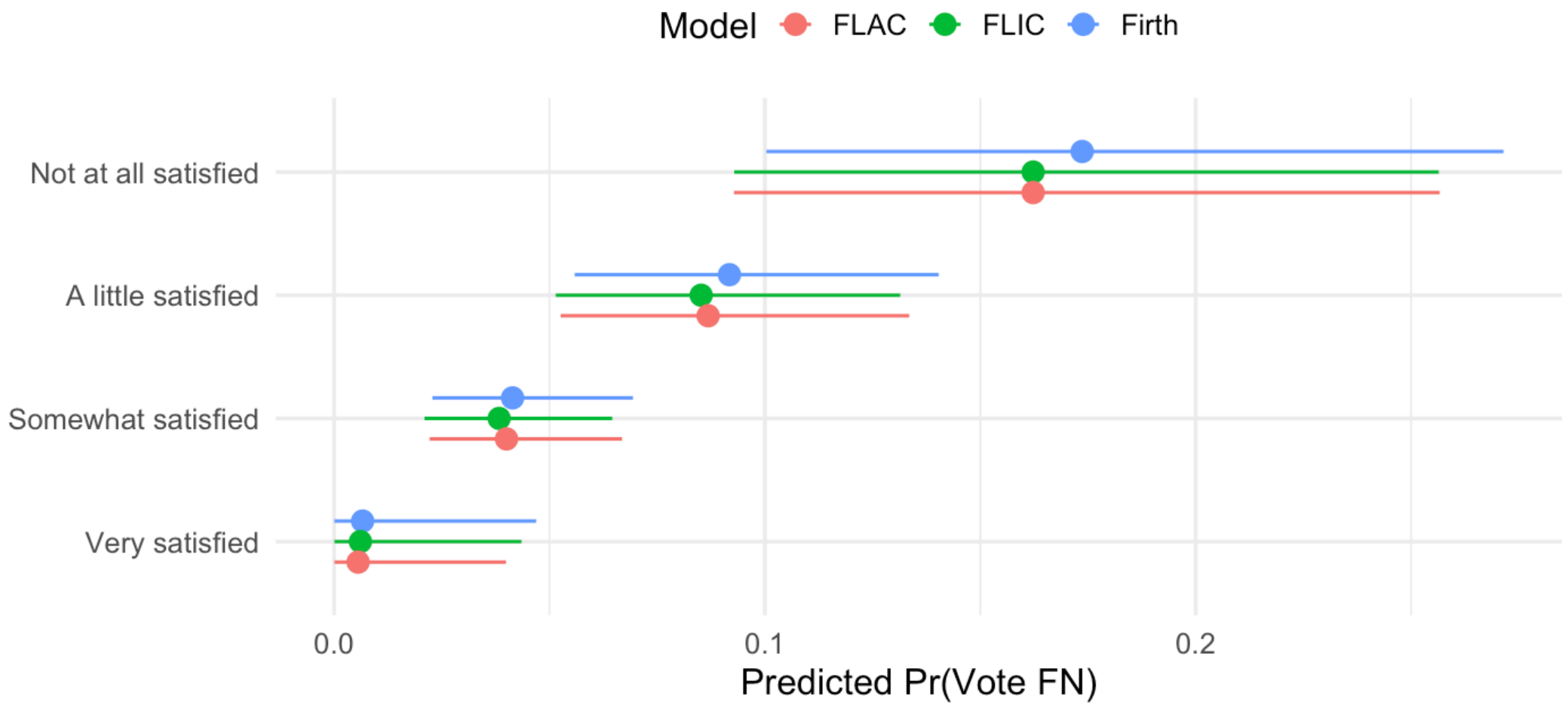
1. Fix the coefficient of interest at a candidate value b.
2. Re-maximize the (penalized) likelihood over all the *other* parameters: $\ell_p(b)$.
3. Compute the deviance from the maximum: $D(b) = 2\left[\ell(\hat{\boldsymbol{\beta}}) - \ell_p(b)\right]$.
4. The 95% CI is every b with $D(b) \leq \chi^2_{1,.95} = 3.84$.

This inverts a likelihood ratio test of $H_0 : \beta = b$. When the likelihood isn't quadratic (separation, small samples), the interval is asymmetric. The same logic works for predicted probabilities and first differences.

Profile vs. Wald: Somewhat Satisfied



Example


Model
FLAC
FLIC
Firth
Not at all satisfied
A little satisfied
Somewhat satisfied
Very satisfied
0.0
0.1
0.2
Predicted Pr(Vote FN)

First Differences

First differences are sometimes less problematic

```
## # A tibble: 6 × 5
##   estimate  lower   upper type    model
##      <dbl>  <dbl>   <dbl> <chr>   <chr>
## 1   -0.167 -0.253 -0.0809 Wald    Firth
## 2   -0.156 -0.241 -0.0716 Wald    FLIC
## 3   -0.157 -0.240 -0.0737 Wald    FLAC
## 4   -0.167 -0.266 -0.0895 Profile Firth
## 5   -0.156 -0.251 -0.0831 Profile FLIC
## 6   -0.157 -0.252 -0.0847 Profile FLAC
```

Review

1. Absolute Measures of Model Fit
2. Comparative Measures of Model Fit
3. Nested and Non-nested Model Tests
4. Model Specification Tests
5. Model Diagnostics
6. (Quasi-)Separation